

Strategising Cloud Migration for Enterprises

Migrating workloads to the cloud requires organisations to consider and carefully weigh factors such as cost, performance and security. This article elaborates on how considering these factors can affect the cloud migration strategy adopted by organisations.



Cloud migration is essentially the process of moving an organisation's applications, data and other workloads to the cloud. Such migration can also be from one cloud to another or a reverse process of cloud repatriation or cloud exit.

In most cases, organisations initiate cloud migration from their data centres to the public cloud so as to take advantage of the elasticity, scalability, redundancy, self-service and the pay-per-use model that the latter offers. There could also be other business drivers that initiate cloud migration – for example, the closing down of a data centre may force an organisation to move workloads quickly to the cloud. In any of these cases, it is important to strategise the move so that factors such as cost, performance and security are considered and optimised.

Planning cloud migration

In order to move workloads to the cloud, the first and foremost thing that an organisation should do is to identify the applications and data that are candidates for the move, because not all of these can be considered for migration. For example, business-critical applications, throughput and latency-centric workloads, applications and data that have strict geographical stewardship such as GDPR, and highly secure data like financial and health records shouldn't be considered for cloud migration.

Next, the right cloud environment should be identified. Questions to be considered when weighing the cost, performance and security considerations for cloud migration are:

- Should the workloads reside in a private cloud?
- Should they be moved to the public

cloud where users can access compute resources via the Internet or a dedicated connection?

- Should workloads be distributed in a hybrid-cloud model which is a mix of private and public cloud?
- Should a multi-cloud environment be used to combine IaaS resources from more than one public cloud provider?

The other consideration to be made by the organisation doing the cloud migration is related to cloud management, for which its staff should be trained.

Types of cloud migration

Cloud migrations are classified based on the way organisations want to use the cloud to accomplish their goals. Apart from differentiating cloud migration based on the degree of adoption (whether it is complete

